

ROBOTICS FOOTBALL GAME

Robo Football is an exciting robotics event where teams design and build their own self-made robots to play football using a soft sponge ball. Each team can participate with one or two robots—one acting as an attacker and the other as a defender if desired. The objective is to score goals while preventing the opponent from scoring. The first team to score 3 goals or the team with the most goals after 20 minutes wins. The event follows an elimination format, combining engineering skill, creativity, and strategy in a thrilling game of robotic football.

Eligibility Criteria

- Open to all students from any department and college.
- Participation is team-based.
- Prior registration is mandatory.
- Basic knowledge/skills related to the event is recommended but not mandatory.

Rules & Regulations

Participation Guidelines

- Team size: 4 members per team.
- Only approved tools, instruments, or resources may be used.
- Prohibited items or practices are not allowed.
- All participants must follow safety and discipline protocols.

Robot Requirements

- Teams may use one or two robots (optional second bot as a defender).
- Robots must be fully self-built by the participants.
- Fully purchased or pre-assembled kits are not allowed.
- Robots must be fully remote-controlled (wired or wireless).
- Maximum robot dimensions: 30 cm × 30 cm × 30 cm.
- Maximum operating voltage: 24V.
- Power source: Batteries only (no fuel-based engines). Batteries must be securely mounted and protected. No open-cell or leaking batteries.

Match / Performance Rules

- Matches are played between two teams.
- Each team may use one or two robots.
- Match duration: 20 minutes (10 minutes per half).
- First team to score 3 goals or the team with the most goals wins.
- Only organizer-provided soft sponge balls will be used.

- Robots must be capable of kicking, pushing, or dribbling the ball.
- No manual interference except through the remote controller.
- Knockout format: Qualifiers → Quarter Finals → Semi Finals → Final → Champion.

Fair Play and Safety

- No hazardous materials, sharp edges, or destructive mechanisms.
- Intentional damage or blocking of opponent robots leads to disqualification.
- Teams must ensure safe wiring and stable robot control.

Disqualification Factors

- Exceeding defined limits.
- Using prohibited items or unfair practices.
- Intentional rule violations.
- Unsportsmanlike behavior or misconduct.

Judging Criteria

- First team to score 3 goals wins.
- If time expires, the team with the most goals wins.
- Judges may consider control, stability, and design innovation in case of a tie.

Game Arena Specifications

- Arena Size: 3500 mm × 2700 mm.
- Boundary Wall Height: 300 mm.
- Goal Post Size: 500 mm × 300 mm.
- Playing Surface: Smooth and flat.
- Ball Type: Soft sponge ball.
- 220V, 50 Hz AC provided only for charging.

Football Game Arena Layout

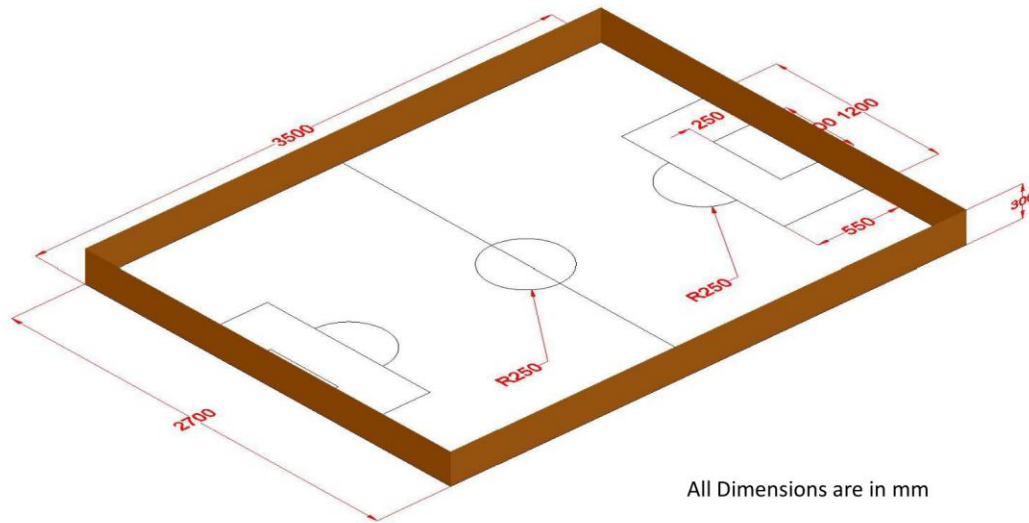


Figure: Official football game arena layout (all dimensions are in millimetres).